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BUILDING BLOCKS FOR FINANCING DEFORESTATION-FREE SUPPLY CHAINS (SMALLHOLDER FOCUS)



This briefing aims to present the essential building blocks to establish a strong, resilient environment that supports deforestation-free supply chains and to bring smallholder farmers to the center of the discussion, guaranteeing their participation in the green transition.

Smallholders are the backbone of global commodity supply chains. Studies suggest that five out of every six farms worldwide are smaller than two hectares, yet they produce up to 35% of the world's food while using only about 12% of global farmland¹. Despite their critical role, they face systemic barriers to financing: lack of collateral, credit history, and formal land documentation often exclude them from traditional banking systems². Low farm-level incomes and limited access to finance restrict smallholders' ability to invest in productivity and sustainability, while concerns about the long-term viability of farming reduce incentives to take on longer-term investments. In many remote regions, weak infrastructure and limited market access further reinforce these constraints. These challenges are compounded by price volatility, climate risks, and inadequate facilities, trapping smallholders in low-income cycles despite their central role in global trade. The EU Deforestation Regulation (EUDR) adds new complexity: while the legal obligation falls on operators and traders placing products on the EU market, compliance depends on geolocation data, traceability, and evidence of deforestation-free production, requirements that often have to be generated upstream and can be difficult to obtain from smallholders who lack resources or technology³. Without targeted support, millions risk exclusion from EU markets, threatening livelihoods and supply chain resilience.

Such support presumes two separate financing perspectives. The first relates to supporting smallholders to transition away from unsustainable agricultural practices by adopting practices, such as agroforestry, improving soil health, and establishing traceability systems. The second one concerns financing smallholders, who are already practicing sustainable farming. Here, investment is needed not only to improve productivity and quality, but also to strengthen resilience and ensure stable, reliable incomes (e.g., through guaranteed income models), so sustainable smallholders can remain competitive in more regulated markets.

To incorporate both of these scenarios, we have elaborated an overarching approach, based on the “Building Blocks” framework. By itself, the Building Blocks framework is not new and, particularly, has been used in research related to “green” transition, and sustainability of agricultural practices. However, it was mainly applied in broader contexts of the system transformation (e.g., “Transformative Change”⁴, “Smallholder value chains as complex adaptive systems”⁵) or tailored to very specific sector contexts. At the same time, it has been less extensively used in the literature to address the financing gap challenge, while we consider it as the core analytical recommendation to leverage smallholder-oriented financial flows.

To this end, we adapt the “Building Blocks” framework to employ ten modular components designed in combination to unlock sustainable smallholder finance. Universal in its components yet adaptable through design efforts, the framework promotes cooperation among farmers, banks, governments, and NGOs to develop solutions suited to local contexts by leveraging financial innovations such as blended finance, risk-sharing mechanisms, and digital tools. By fostering multi-stakeholder partnerships, smallholder finance can shift from a bottleneck to a driver for EUDR-aligned, deforestation-free value chains.

1 [FAO \(2021\)](#)

2 [World Bank \(2022\)](#)

3 [Regeneration \(n.d.\)](#)

4 [International Climate Initiative \(2020\)](#)

5 [Alliance \(2015\)](#)

BUILDING BLOCKS

The Building Blocks are fundamental, interchangeable components that, together, establish a comprehensive system to promote sustainable smallholder financing. They serve a dual purpose.

- **Structural Role:** Defining the architecture of a resilient financing ecosystem.
- **Transformational Role:** Specific methods to catalyze systemic change when needed.

By organizing efforts around these Building Blocks, stakeholders can strengthen their focus on what content to address and how to manage it efficiently, whether the aim is to empower existing smallholder systems or drive transformation. Although definitions and classifications may differ, there is strong agreement on the focus areas of change, making the Building Blocks universal yet adaptable across diverse actors and regional contexts.

METHODOLOGY

The framework consists of ten building blocks, structured in two parts: the Foundation and the Roof. Importantly, building blocks differ from mere requirements or conditions in that they are enabling rather than restrictive, composable rather than binary, and architectural rather than regulatory. This means that each building block can possibly generate multiple context-specific requirements, which may also evolve over time.

FOUNDATION

Seven building blocks establish a solid foundation, supporting the remaining higher-level building blocks. These foundational building blocks are logically interconnected among each other, as conceptually, they are meant to work together to align incentives of actors along the value chain, as well as diverse stakeholders.

Specifically, seven building blocks were designed to address the three guiding questions (WHY, HOW, and WHAT):

- **WHY** must stakeholders act: cost reduction (**BB2**) and improved livelihoods (**BB3**)
- **HOW** to implement sustainable practices: being regulatory-compliant (**BB1**) and having a shared vision (**BB4**)
- **WHAT** constitutes and allows for a sustainable practice: no capital flow barriers (**BB5**), no data flow barriers (**BB7**) and leveraging innovative financial instruments and mechanisms (**BB6**)

Building Block	Explanation
BB1 - Regulatory and Policy compliance	Systems enabling progressive alignment of smallholder production with deforestation-free, EUDR - compliance
BB2 - Access to Technology	Digital and technological infrastructure for targeted financing for smallholders
BB3 - Socio-economy	Socio-economic enabling environment and constraints (aspects of a location allowing sustainable agri-practices)
BB4 - Sustainability mindset	Stakeholders' mindset empowering smallholders' sustainable agri-practices
BB5 - Available Financial flows	Financial connectivity among stakeholders and risk-sharing capital structures
BB6 - Innovative financial tools	Access to innovative financial instruments and mechanisms
BB7 - Available Information flows	Data systems enabling information sharing and trust among stakeholders

Besides, the same seven building blocks reflect two mirrored priorities:

- **Efficient Allocation of Funds:** BB1, BB2, BB5, BB6
- **Efficient Use of Funds:** BB3, BB4, BB6, BB7

These sides stand for two essential pillars for developing a robust funding environment for deforestation-free supply chains and are both leveraged through a common component of financial innovations (BB6).

Finally, foundational building blocks integrate three dimensions, forming a holistic approach to sustainability:

- **Environmental stewardship:** BB1, BB4
- **Technological innovation:** BB2
- **Social equity and economic viability:** BB3

ROOF

The top layer of the Building Blocks structure represents accelerators of transformation and overarching conditions for success.

Building Block	Explanation
BB8 – Capacity Building	Developing stakeholders' capabilities to implement and disclose sustainable practices
BB9 – Awareness-raising	Promoting importance and successful examples of sustainable practices
BB10 - Stakeholder cooperation	Mechanisms enabling collective stakeholder action, mutual responsibility

These elements serve as catalysts, enabling core components and customizing them across various contexts. Together, the Foundation and Roof form a comprehensive, systemic, and practical approach to address challenges in funding deforestation-free supply chains and open related opportunities.

Discussions among various stakeholder groups to promote sustainable finance for smallholders can be organized around the same Building Blocks, while allowing for tailored design efforts and topics depending on participants and regional context.

FINANCING BARRIERS AND OPPORTUNITIES ACROSS STAKEHOLDERS

This section explores how the Building Blocks framework operates in practice: it exemplifies challenges and opportunities faced by key stakeholder groups towards creating a resilient environment that enables smallholder farmers to thrive in deforestation-free supply chains.

The analysis identifies the barriers and how they can become catalyst actions, unlocking capital for sustainable and inclusive supply chains, by mapping the relevant Building Blocks for each stakeholder:

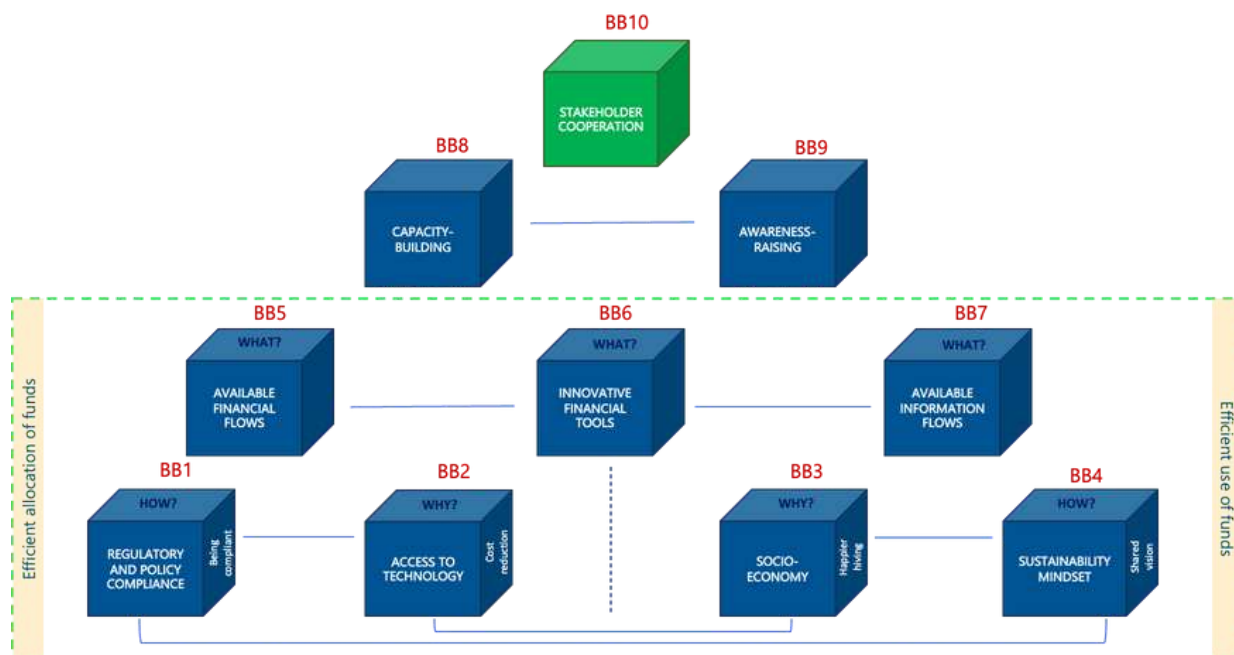


Figure 1 - Building Blocks' structure

The Building Blocks are relevant to different stakeholders; they present challenges for each to address and, at the same time, offer opportunities once they are in place.

Stakeholder	Challenge	Opportunity (to be leveraged through joint action)	Building Blocks									
			1	2	3	4	5	6	7	8	9	10
Smallholders	- Limited participation in cooperatives and lack of awareness about partnerships strategies - High Capital Expenditure (CAPEX) and pre-financing needs - High insurance premiums due to poor risk data - Lack of incentives for sustainability	- Strengthen cooperatives for bargaining power - Promote Good Agricultural Practices (GAP) and success stories - Develop pooled procurement systems - Introduce parametric insurance and blended finance - Offer premium pricing and performance-based finance (e.g., carbon credits-where viable)			X	X	X	X	X	X	X	X
Banks	- Limited financial literacy among smallholders - Distorted pricing due to poor data - High default risk from climate volatility - Lack of innovative risk-sharing tools	- Develop credit guarantees and first-loss capital - Improve data collection for risk pricing - Create blended finance and impact metrics - Diversify borrower portfolios	X		X	X	X	X	X	X	X	X
Government (local & national)	- Lack of credit guarantees and legal frameworks - Gender inequality and declining rural economy - Limited capacity for resilience planning - Fragmented coordination among sectors	- Provide blended finance and co-financing schemes - Promote social equity and guaranteed income models - Facilitate water management and tech access - Encourage sector cooperation to address small ticket sizes	X	X	X	X	X	X	X	X	X	X
Regulatory Authorities	- High certification costs and complex compliance - Privacy vs traceability trade-offs - Lack of harmonized reporting standards - Risk of exclusion due to rigid EUDR requirements	- Simplify compliance and reporting processes - Introduce incentives for traceability (premiums) - Develop solutions for data governance and privacy - Enable holistic view of sector impacts	X			X	X	X	X		X	X
Foundations/ Philanthropies	- Weak governance for risk-sharing - Limited advisory capacity for climate-smart farming - Sensitivity to interest rates and cost shocks - Lack of accountability mechanisms	- Enable robust governance frameworks - Provide blended finance and long-term commitments - Fund literacy programs and climate-smart practices - Support socially equitable and commercially scalable solutions			X	X	X	X	X	X	X	X

The building blocks can be utilized in **transformational** or **strengthening** scenarios; the former possesses the greatest potential, as it predominantly occurs in developing countries and could yield the most significant benefits. In that sense, we analyzed the Uganda Coffee sector.

Uganda's coffee sector provides a high-impact reference case for understanding the systemic challenges and opportunities in creating deforestation-free supply chains. Coffee is Uganda's leading agricultural export, contributing up to 20–30% of foreign exchange earnings and supporting over 1.7 million smallholder households⁶. However, the sector faces fragmented cooperative structures, limited access to finance, gender inequality, and climate vulnerability—issues that mirror the stakeholder challenges identified earlier. Evidence from Uganda's coffee-growing regions shows that farmers are already experiencing longer droughts and more erratic rainfall, which negatively affects flowering and bean filling, alongside increasing pressure from pests and diseases⁷. These barriers not only constrain productivity but also heighten the risk of unsustainable practices, making Uganda an ideal example to demonstrate how the Building Blocks framework can transform risks into opportunities through coordinated action.

Gender inequality further compounds these challenges. In Kanungu district, women provide most coffee labour—58% during fieldwork/harvest and 72% in post-harvest handling—yet men largely control marketing, processing, and income from sales⁸. Meanwhile, meeting emerging sustainability requirements such as the EUDR is essential to retain access to premium markets, which make up over 60% of Uganda's coffee exports⁶. Together, these dynamics highlight the need for transformational action beyond incremental change.

Building Blocks		Role in Uganda's Coffee Sector	Priority Actions (collective responsibilities)
BB1	Regulatory & Policy Compliance	Harmonise sustainability standards and simplify certification to reduce costs for smallholders.	Develop national deforestation-free guidelines; streamline certification processes.
BB2	Access to Technology	Enable traceability and climate-smart practices through digital tools.	Invest in mobile platforms for risk data and compliance tracking.
BB3	Socio-Economy	Address gender inequality and rural poverty to stabilise household economies.	Promote income-stabilisation mechanisms (e.g., minimum price guarantees/price floors, longer-term offtake contracts and timely payments) and gender-inclusive programmes
BB4	Sustainability Mindset	Foster awareness of GAP and climate-smart practices among farmers.	Implement training and awareness campaigns at cooperative level.
BB5	Available Financial Flows	Mobilise affordable credit and blended finance for smallholders.	Set up blended-finance facilities offering seasonal working capital (inputs/pre-harvest finance), equipment leasing, and partial credit guarantees via cooperatives/aggregators.
BB6	Innovative Financial Tools	Reduce climate and market risks, and transaction costs, through risk-sharing and cash-flow tools (beyond credit).	Deploy a mix of risk-sharing tools (e.g., weather-index/parametric insurance, guarantee/first-loss facilities, and price-risk tools such as forward contracts) linked to data and pooled finance.
BB7	Available Information Flows	Improve risk assessment and traceability through reliable data systems.	Establish a shared sector data platform (hosted by a neutral entity, e.g., coffee authority/industry body/trusted third party) with clear governance, data ownership/consent, and access rules.
BB8	Capacity-Building	Strengthen cooperatives and local institutions for governance and negotiation.	Provide technical and managerial training for farmer organisations.
BB9	Awareness-Raising	Shift perceptions toward sustainability and showcase success stories.	Launch campaigns highlighting benefits of deforestation-free coffee.
BB10	Stakeholder Cooperation	Align incentives and scale solutions through multi-stakeholder platforms.	Establish sector-wide coordination bodies involving farmers, banks, and regulators.

⁶ [Uganda Coffee Platform \(2024\)](#).

⁷ [Jassogne, Laderach & van Asten \(2012\)](#).

⁸ [Farm Africa \(2020\)](#).

This integrated approach ensures the creation of an enabling environment that unlocks capital and promotes transparency, driving scalability and resilience for smallholders in the country. To foster this environment, all the stakeholders should be involved in the implementation.

The leadership of this development will depend on the desired outcome. If the main goal is to implement a national monitoring system, the Local government should oversee it. To secure socioenvironmental safeguards for smallholder farmers, regulatory authorities should lead this development.

On the other hand, if the goal is to create innovative and inclusive financial products, Band and financial institutions should spearhead them. Despite these initiatives having a single stakeholder in charge of their development, all actors should participate in and contribute to their planning and implementation.

Uganda's coffee sector can thus serve as a blueprint for other commodity systems, demonstrating how multi-stakeholder collaboration and innovative financing can deliver sustainable, deforestation-free supply chains.

KEY TAKEAWAYS

The shift towards deforestation-free supply chains places smallholders at the heart of global sustainability efforts. Yet, systemic barriers, such as a lack of financing, technical capacity, and regulatory complexity, threaten their inclusion. The Building Blocks framework provides a framework for addressing these challenges by combining financial innovation, capacity-building, and stakeholder collaboration. Success relies on integrated approaches that align incentives, simplify processes, and tailor solutions to local contexts, ensuring that smallholder finance moves from being a bottleneck to a driver of transformation. The five key takeaways, and some concrete examples, can be found below:

- **Collaboration and capacity-building are essential:** Multi-stakeholder cooperation (BB10) and capacity-building (BB8) consistently accelerate systemic change.
 - SAFE Sustainable Finance Knowledge Journeys: Online sessions to build capacities on how development cooperation, NGOs, Think Tanks, private sector, International Organizations, etc. are working together to close smallholder finance gaps,
- **Financial innovation and regulatory alignment are critical:** blended finance, risk-sharing mechanisms (BB6), and clear compliance frameworks (BB1) unlock sustainable finance to finance EUDR readiness.
 - IDH's Farmfit and PPI models, Brazilian rural credit (PRONAF/SNCR), IFC/ World Bank guarantees for agri-SME. Check also this Financial Solutions Database, an open-access tool that catalogues real financing mechanisms to support smallholder-inclusive, EUDR-aligned, deforestation-free supply chains.
- **Data and awareness matter:** Improving information flows (BB7) and raising awareness (BB9) complement technical and financial measures, empowering smallholders to participate in sustainable value chains.
 - National cocoa traceability systems in Ghana and Côte d'Ivoire (and Cameroon-in progress), Proforest traceability & smallholder engagement.

- **Socio-economic focus enables win-win:** Addressing socio-economic concerns of particular communities (BB3), helped by more technologically efficient allocation of funds (BB2) and shared benefits of sustainability (BB4), importantly contribute to removing barriers in financial flows to smallholders (BB5).
 - (Proforest and GIZ have their own advances in landscape and jurisdictional approaches. However, they have also joined forces with Nestlé & partners to form a multi-stakeholder landscape initiative (SUSTAIN KUTIM, Indonesia) aligning companies, local government and communities to protect forests while supporting smallholders in palm oil and rubber supply chains.
- **Customisation drives impact:** Adapting Building Blocks to specific stakeholder needs, regional and sector realities, and transformation scenarios ensures practical implementation and scalability.
 - Indonesia & Malaysia: smallholder-first customization for palm/rubber/cocoa—diagnose constraints, align ISPO/MSPO with EUDR, and scale traceability + finance

➤➤➤ CONTACT INFORMATION:

Elisabeth Hock (elisabeth@climcom.org)
 Enzo Bernardinelli (enzo.bernardinelli@climcom.org)
 Kristina Lyadskaya (kristina.lyadskaya@climcom.org)
 Paula Zambrano (paula@climcom.org)

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